

More Linear Programming

Linear programming can be used to solve minimization problems.

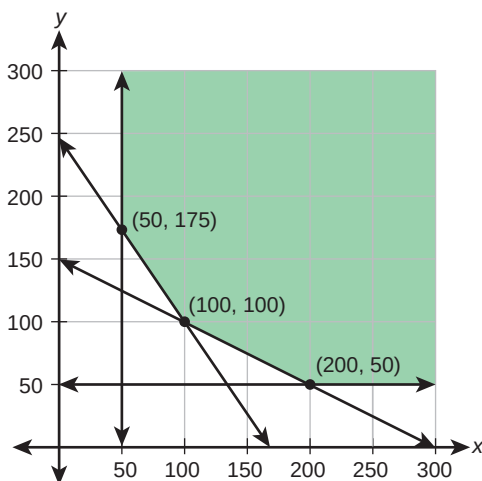
EXAMPLE

Minimize $C = 7.50x + 5.50y$ subject to the following constraints:

$$\begin{cases} 3x + 2y \geq 500 \\ x + 2y \geq 300 \\ x \geq 50 \\ y \geq 50 \end{cases}$$

SOLUTION

Graph the constraints to construct the feasible region. Then find the vertex points.



The vertex points are (50, 175), (100, 100), and (200, 50).

Evaluate the objective function at each vertex point.

Vertex point	$7.50x + 5.50y$	C
$(50, 175)$	$7.50 \cdot 50 + 5.50 \cdot 175$	1337.50
$(100, 100)$	$7.50 \cdot 100 + 5.50 \cdot 100$	1300
$(200, 50)$	$7.50 \cdot 200 + 5.50 \cdot 50$	1775

Since the value of C is least at $(100, 100)$, the minimum of the objective function is 1300. ■